Computer Project 3: Continuum Kinetic Plasma Simulations

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In this paper, Vlasov's equation is solved numerically to perform continuum kinetic plasma simulations. The computations are performed on an 1D1V grid on which contributions of the magnetic field are neglected so that Maxwell's equations reduce to solving Poisson's equation. The partial differential equation is solved by decomposing it into two 1D stencils that are combined using Strang splitting. Both dimensions are discretized by using an upwind method that combines forward Euler in time with the correct one-sided difference depending on the advection speed in the respective dimension. The numerical simulation is used to study weak and strong Landau damping, as well as the two-stream instability. Despite the simple numerical implementation, all cases produce results that resolve the physical processes correctly. Still, the low numerical accuracy leads to characteristic growth rates that, in parts, differ strongly from the theoretical values.

INTRODUCTION

In general, plasmas are systems of separated charges that exhibit collective behavior and quasi-neutrality on macroscopic scales. A first approach to model these systems numerically could be to describe the system in a Lagrangian reference frame. For this approach, the plasma is described by distinct particles that each have their own position and momentum variables, as well as a specific charge. All charges couple to Maxwell's equations which lead to the fields that determine the forces acting on the particles. With this method, the plasma could be modeled as a closed system without the need for simplifying assumptions. However, the large number of particles and even larger number of particle interactions make this model impractical due to computation times exceeding any acceptable scale. Instead, reduced kinetic methods are used that either reduce the number of particles by forming super particles (Particle-In-Cell) or describe the phase-space by distribution functions (Continuum Kinetic Models). The latter of which will be explored in this paper.

Continuum kinetic models perform an ensemble average over the phase-space leading to a smooth description of the discrete spectrum. The resulting probability distribution function assigns a probability value to every point in the phase-space; an inherently Eulerian description. The dynamics are then described by the Boltzmann equation for a system that only interacts via the Lorentz force that takes the form

$$\frac{\partial f_s}{\partial t} + v \cdot \frac{\partial f_s}{\partial \boldsymbol{x}} + \frac{q_s}{m_s}\big(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\big) \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} = \frac{\partial f_s}{\partial t}\bigg|_{\text{coll}}. \quad (1)$$

Here, f_s is the probability distribution function of a given species of particles s with charge q_s, mass m_s, reacting to an electric field $\boldsymbol{E}$ and a magnetic field $\boldsymbol{B}$. The term of the right-hand side of the equation refers to the temporal change of the distribution function due to collisions. The distribution function itself couples to Maxwell's equations by the charge and current density imposed on the space domain by

$$\rho_c(\boldsymbol{x}) = \sum_s \int \mathrm{d}\boldsymbol{v}\; q_s f_s(\boldsymbol{x}, \boldsymbol{v}), \quad (2)$$

$$\boldsymbol{j}(\boldsymbol{x}) = \sum_s \int \mathrm{d}\boldsymbol{v}\; q_s \boldsymbol{v}_s f_s(\boldsymbol{x}, \boldsymbol{v}). \quad (3)$$

The continuum kinetic model describes the evolution of plasmas very accurately but requires a lot of computational power since the distribution function is a 6D field. Thus, this method is mostly used to study phenomena with small spatial and temporal scales, like wave-plasma interactions. Additionally, for processes where the tails of the probability distribution are especially relevant (like for fusion of ionization), a description with the continuum kinetic model is worth the additional computational effort.

In this paper, the power of the continuum kinetic model is studied by using a very simple domain with one spatial and one velocity dimension (1D1V). Furthermore, the magnetic field is neglected so that Maxwell's equations can be reduced to solving Poisson's equation. In this domain two very common plasma processes are seeded to test the accuracy of the employed numerical method. On one hand, the interaction between the plasma and a single plasma wave is studied for small and large wave amplitudes, to trigger weak and strong Landau damping. On the other hand, the domain is truncated with two Maxwellian peaks in the distribution function (in the velocity space) creating two opposing streams. Additionally, a small wave perturbation in space is added that feeds from the streams and grows exponentially in amplitude. This is the two stream instability.

CONTINUUM KINETIC SIMULATIONS

Vlasov-Poisson Solver

For the numerical simulation, the plasma is assumed to be collision-less, meaning that the collision term on the right-hand side is zero. Furthermore, the ions are approximated as a constant homogenous space charge distribution. This is a good approximation since the ion mass is much bigger than the electron mass so that the Lorentz force will cause almost no ion acceleration. Thus, the system is fully described by the so-called Vlasov equation for electrons in 1D1V (since this is the only species, the subscript is omitted):

$$\frac{\partial f}{\partial t} + v \cdot \frac{\partial f}{\partial x} + \frac{e}{m}\big(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\big) \cdot \frac{\partial f}{\partial v} = 0. \tag{4}$$

To simplify the numerics, the electrons charge and mass are set to one ($e = m = 1$).

In a first attempt, Vlasov's equation was discretized following a Lax-Friedrichs scheme. After a simultaneous 2D step for both dimensions led to high numerical diffusion, the time advance was sub-divided using Strang splitting, in an attempt to decrease the diffusion error. For Strang splitting, every time advance is separated into three 1D time steps: one half step for the space flux, one full step for the velocity flux, and finally another half step for the space flux. The maximum time step for this scheme is determined through separate Von Neumann stability analysis of the two 1D steps so that

$$\Delta t \leq \min\left(\frac{\Delta x}{\max(v_j)}, \frac{\Delta v}{\max\left(\frac{eE_i}{m}\right)}\right). \tag{5}$$

However, the Strang splitting alone didn't lower the numerical diffusion sufficiently. Thus, it was decided to switch from the Lax-Friedrichs method, to a simple upwind method that uses forward Euler in time and a one-sided difference in space. Since the one-sided derivative is only stable for advection velocities in the respective direction, the direction is adaptively chosen according to the sign of the advection velocity. The Strang splitting is kept, so that the advection velocity in position space is v_i, while the advection speed in velocity space is given by $a = eE_i/m$. The constraint on Δt remains the same as in equation 5.

Since the magnetic field is neglected, the field resulting from the charge distribution is calculated with Poisson's equation for the electric potential Φ

$$\nabla^2\Phi = -\rho_c/\epsilon_0, \tag{6}$$

that relates to the electric field strength with $\boldsymbol{E} = \nabla\Phi$. Again, $\epsilon_0 = 1$.

Numerically, the elliptic Poisson equations is solved by expressing it in the form of a linear system $\mathrm{A}\boldsymbol{x} = \boldsymbol{b}$, where the Laplace operator is matrix A and the source term is given by the charge density ρ_c that can be calculated by integrating the probability distribution function over the velocity according to equation 2. Since the laplacian doesn't change over time, it is computed once at initialization and then repeatably utilized to solve the linear system after every full time step.

For the boundary conditions periodicity is assumed for x, while zero flux is required for both boundaries of v. In the code, the Lax-Friedrichs steps are implemented by rolling the arrays to the side, so that the periodic x boundary condition is fulfilled trivially. For the velocity, two additional ghost cells are added on both sides that are updated with the adjacent interior value after every time step. This keeps the boundary flux at zero.

All in all, the code is modularized into four different classes. First, the *FDMGrid* class holds all information about the finite difference grid and contains the methods that are used to enforce the boundary conditions or initialize the simulation domain. Next, the *VlasovPoissonSolver* class contains the numerical discretization logic and defines the full time step for the staged *Lax-Friedrichs* scheme. Further, the Simulator class handles the simulation procedure, logs the electric field energy over time, and also saves snapshots of the simulation domain. Finally, the *DataAnalysis* class is used to analyze the simulation data by creating plots and performing linear fits to the logarithmic field energy evolution.

Landau Damping

Landau damping describes the damping process of longitudinal space charge waves in plasmas. In other words, a space charge oscillation that travels through a conductive fluid (i.e. a plasma) undergoes an exponential decay of energy for small amplitude variations. It is important to note that this effect also exists in collision-less plasmas. Thus, the energy is not dissipated through heating due to collisions but rather transfers energy from the electric field directly into the velocity distribution.

For large perturbation amplitudes, the effect becomes non-linear and the energy in the electric field starts to grow exponentially after an initial exponential decay. This happens because the plasma particles get trapped in the fields potential and start oscillating together with the initial perturbation. The oscillation causes an energy transfer from the velocity distribution into the field.

The initial distribution function and the initial electric field are chosen according to a reference paper [1], to make the simulation results comparable. Thus, the probability distribution function is initialized with a Maxwellian in velocity space and a sinusoidal variation in amplitude along the position space that takes the form

$$f(x,v) = \frac{n_0}{v_{\mathrm{th}}\sqrt{2\pi}} \exp\left(-\frac{v^2}{2v_{\mathrm{th}}^2}\right)\left(1+\alpha\cos(kx)\right). \quad (7)$$

Here, n_0 is the number density of the homogenous proton background, v_{th} is the thermal velocity, α is a scaling parameter for the amplitude of the perturbation, and k is the wave number of the perturbation in position space. The corresponding electric field is given by

$$E(x) = n_0 \frac{\alpha}{k} \sin(kx). \quad (8)$$

This initial state is then propagated through time using the simulation setup described above. The results are then visualized by two separate plots. One shows the evolution of the electric fields energy over time, while the other depicts the distribution function at a point in time at which the damping effect is clearly visible. This procedure is performed for the weak damping case ($\alpha = 0.01$) and the strong damping case ($\alpha = 0.5$).

The case of weak Landau damping shows the expected exponential decay of the electric field energy over time, until the curve flattens out after approximately 40 characteristic periods (see Figure 1). Since the timing coincides with the recurrence in the reference paper it is assumed that the process observed here is connected to this. Due to the first order accuracy in time and space of the employed finite difference method, the initial structure is out of shape by the time recurrence would occur. Therefore, instead of resembling the initial state, the system finds some sort of oscillatory mode that keeps the field energy constant. The low accuracy combined with the small perturbation is probably also the reason why the measured damping factor is around factor 2 smaller than the theoretical value.

Figure 2 shows the distribution function in phase space for weak Landau damping after initialization and towards the end of the exponential decay. Besides some faint horizontal structures that are due to the advection in position space, the distribution almost looks identical. This is expected since the perturbing amplitude in this case is very small.

For the strong Landau damping, both the initial decay and the subsequent exponential growth are clearly visible in Figure 3 and coincide with the timings in [1]. While the simulated decay factor is only about 18 % smaller than the theoretical value, the growth rate almost 70 % smaller than expected. Again, the low accuracy of the discretization is probably the culprit for this discrepancy. The initial decay factor here is probably closer to the expectations than for the weak damping case because the perturbation amplitude is a lot larger and the process thus simpler to resolve.

The distribution function gives a more dynamic picture than for the weak damping case. Figure 4 shows the characteristic horizontal stripes that are expected from the almost circular initial distribution. After the damping period, the distribution has spread out in velocity space, representing the transfer of energy from the electric field into the plasma. After that, the particles are trapped in the potential and collectively transfer energy back into the field. On the final plot that shows the phase space at the end of the growth process, the distribution is more compacted in velocity space again, visualizing this energy transfer. Moreover, the waveform along position space relates to the electric field perturbation showing the coupling between the plasma and the perturbation.

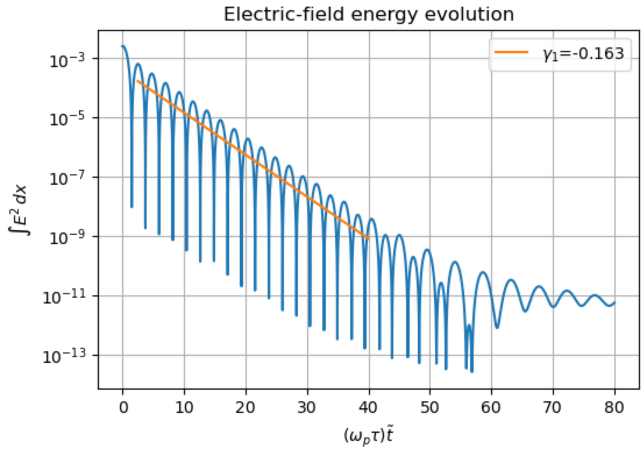

Figure 1. This logarithmic evolution of the energy stored in the electric field depicts the exponential decay in the regime of weak Landau damping. After 40 characteristic periods, the process stops due to an effect related to recurrence and low numerical accuracy.

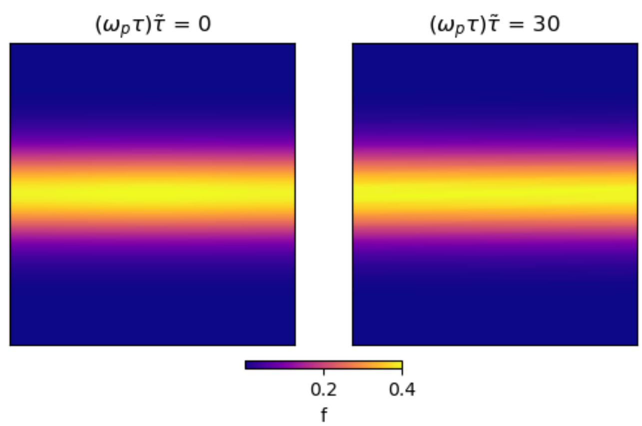

Figure 2. The distribution function is plotted over the whole simulation domain with x-axis $\in [-2\pi, 2\pi]$ and v-axis $\in [-5, 5]$. It does barely change over time for the weak Landau damping case. The small perturbation is not visible on this plot, so the only difference are some horizontal structures that result from the advection in position space.

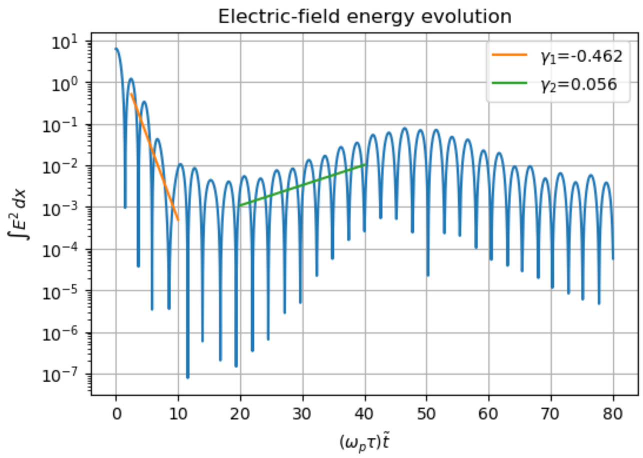

Figure 3. For strong Landau damping, the field energy decays until an exponential growth sets in. The growth factors differ from what is expected from theory but the general process is resolved.

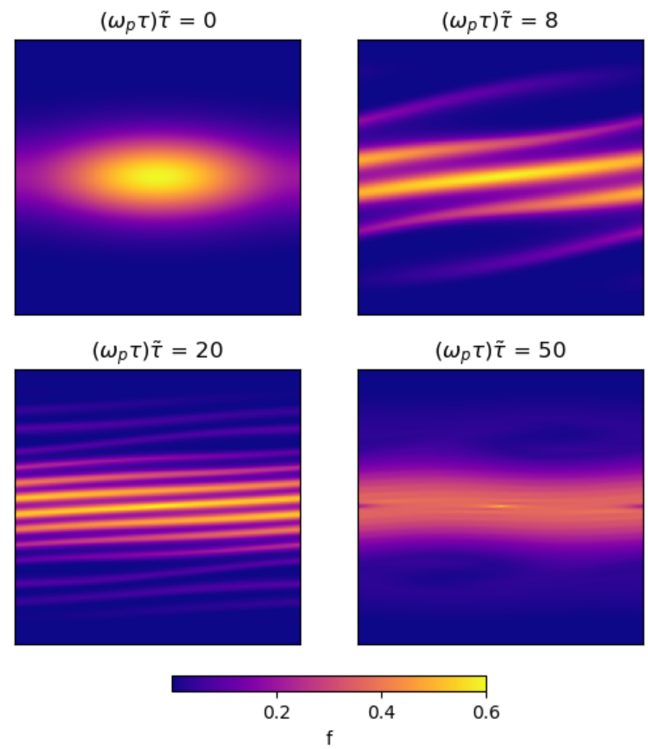

Figure 4. The distribution function is plotted over the whole simulation domain with x-axis $\in [-2\pi, 2\pi]$ and v-axis $\in [-5, 5]$. For the strong Landau damping case, the horizontal structures are now separated because of the large initial perturbation in position space. During the damping of the field, the plasma gains velocity, before it is transferred back into the field during the growth process.

Two-Stream Instability

The two-stream instability describes the exponential growth of a plasma wave that can appear when two beams, or a plasma and a beam are intersected with each other, so that their distribution functions overlap. This leads to a distribution function with two distinct bumps. When the phase velocity of a plasma wave now sits between those two bumps, energy can be transferred from the velocity distribution into the plasma wave, leading to an exponential growth.

To simulate this phenomenon, the initialization is again conducted according to the reference paper [1]. The setup is very similar to the Landau damping case. However, in this case, two separate Maxwell-Boltzmann distributions are introduced in velocity space so that the domain is filled with two opposing streams

$$f(x,v) = \frac{n_0}{v_{\text{th}} 2\sqrt{2\pi}} \left[\exp\left(-\frac{(v-v')^2}{2v_{\text{th}}^2}\right) + \exp\left(-\frac{(v+v')^2}{2v_{\text{th}}^2}\right) \right] \big(1+\alpha\cos(kx)\big). \tag{9}$$

Since the spatial variation of the distribution function has exactly the same sinusoidal relationship as the Landau damping case, the initial electric field is initialized exactly like in equation 8.

The evolution of the field energy for the two-stream instability simulation shows how even a small perturbation with $\alpha = 0.01$ grows exponentially in amplitude (see Figure 6). Again, the general evolution of the results closely match with the reference paper. However, the growth factor is about 37 % smaller than expected from theory. Like above, this can be explained with the low accuracy of the employed numerical method.

Figure 5 illustrates the evolution of the two-stream distribution in phase space. Initially, the two symmetric Maxwellian humps are visible as two well-separated horizontal lines. Due to the small perturbation in position space, the lines slowly start to bend into wave-like structures, while the electric field's energy starts to grow exponentially. The bending is due to acceleration by the electric field. As the instability grows, rings start to form around two of the roots of the electric field, where the particles are now trapped in an oscillatory motion in the potential of the perturbation. The characteristic cat's eye starts to form at saturation, where the wave growth stops. Over time the eye starts to decompose into a broad plateau around $v = 0$. It is remarkable how close the simple numerical method implementation comes to the sophisticated solver employed to create the reference plots in Figure 4 in [1]. The accuracy causes more smearing and diffusion, but the general dynamics are still resolved.

[1] A. Ho, I. A. M. Datta, and U. Shumlak, Physics-based-adaptive plasma model for high-fidelity numerical simulations, Frontiers in Physics **6**, 105 (2018).

Appendix

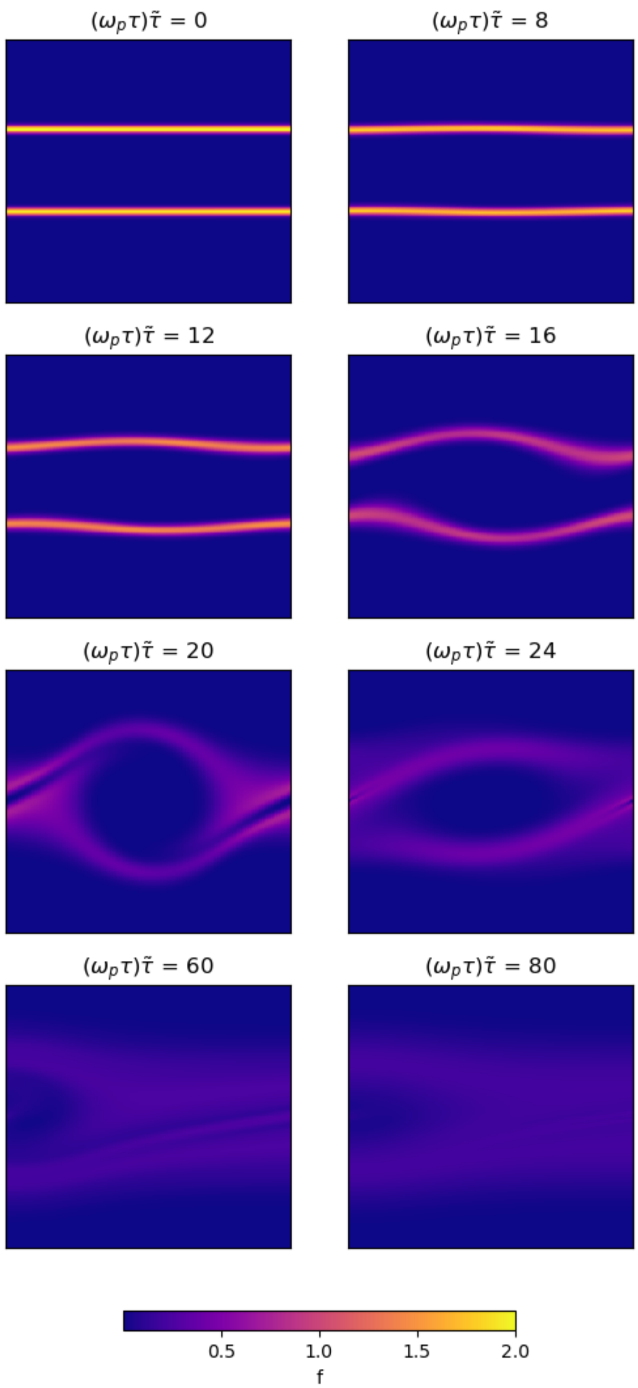

Figure 5. Plot of the distribution function for one half of the simulation domain with x-axis $\in [0, 4\pi]$ and v-axis $\in [-5, 5]$. Growth of the instability occurs at the expected rate until saturation at $(\omega_p\tau)\tilde{\tau} = 20$.

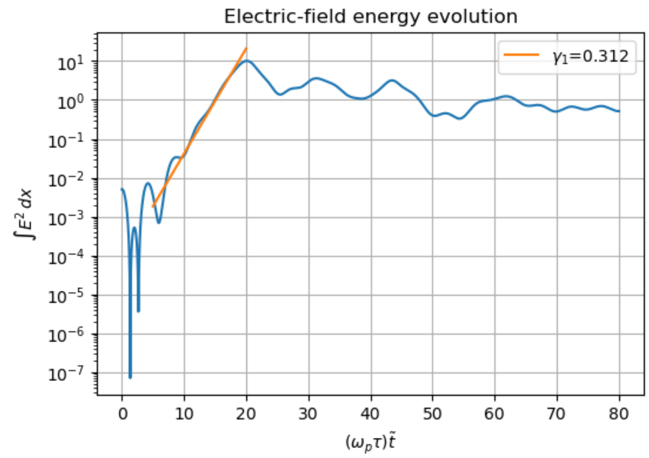

Figure 6. The evolution of the electric field's energy shows an exponential growth until about 20 characteristic periods. This portrays how energy is transferred from the plasma into the electric field.